

# Redshift SOC

## SOC Home Lab — Build Phases

*This documentation is written to be reproducible by others building a similar environment anyone looking to construct a comparable SOC lab for learning or portfolio purposes can follow the phase objectives below as a reference plan*

### Overview

This document lists the phases planned and executed for the Redshift SOC home lab — a segmented, single-laptop Security Operations Center environment built with VMware Workstation Pro, pfSense, Windows and Linux endpoints, Kali Linux, and Splunk Enterprise. Each phase below reflects the objective agreed on at the start of that stage of the build.

### Build Phases

| Phase | Name                                         | Objective                                                                                                                                          | Status   |
|-------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1     | Lab Architecture & Environment Planning      | Design the network topology, choose the hypervisor and firewall platform, and plan resource allocation before any installation.                    | Complete |
| 2     | Hypervisor & Base Network Configuration      | Install VMware Workstation Pro and create the isolated virtual networks (attacker segment, victim segment) that the lab's segmentation depends on. | Complete |
| 3     | pfSense Firewall Deployment                  | Deploy pfSense as a three-interface firewall (WAN / LAN / OPT1) enforcing default-deny between the attacker and victim segments.                   | Complete |
| 4     | Windows Endpoint Deployment                  | Deploy a Windows victim/monitored endpoint on the internal segment with a static IP.                                                               | Complete |
| 5     | Ubuntu Server Deployment                     | Deploy a headless Ubuntu Server as the second victim machine and SSH target.                                                                       | Complete |
| 6     | Kali Attacker Deployment                     | Deploy Kali Linux on the isolated attacker segment and verify network isolation before any firewall rules exist.                                   | Complete |
| 7     | Apache Web Server Deployment                 | Install Apache on the Ubuntu server as the web application target, with verified custom content and access logging.                                | Complete |
| 8     | Firewall Rule Verification & Traffic Logging | Open specific, logged firewall rules for attacker-to-victim traffic and verify both allowed and blocked paths behave correctly.                    | Complete |
| 9     | Splunk Enterprise Deployment                 | Stand up Splunk Enterprise as the lab's centralized log collection and analysis platform.                                                          | Complete |

|    |                                                      |                                                                                                                                                    |             |
|----|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 10 | Log Forwarding (Sysmon, Universal Forwarder, Syslog) | Install Sysmon on Windows and Splunk Universal Forwarders on all endpoints, and forward pfSense syslog, so all log sources feed into Splunk.       | Complete    |
| 11 | Attack Simulation & Detection Validation             | Run safe, MITRE ATT&CK-mapped attack simulations from Kali against the victim endpoints and confirm the resulting telemetry is captured in Splunk. | In Progress |
| 12 | Triarc — AI-Assisted Alert Triage                    | Build an AI-assisted layer that reviews fired alerts and recommends a True/False Positive classification with supporting rationale.                | Planned     |

## Notes

This phase list reflects the original build plan agreed at project start. Detailed implementation steps, configuration values, and troubleshooting for each phase are maintained separately and available on request.